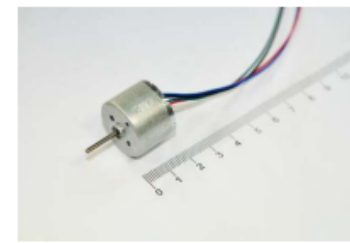


Brushless DC Motor equipped with ball bearings

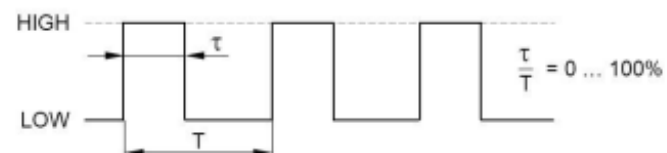


This Electric Drive Specification is only valid for the electric drive itself. The specification may vary in combination with a pump.

Basic specification				
Motor type	2-wire		4-wire	
Nominal supply voltage	12 V	24 V	12 V	24 V
Supply voltage range	8 ... 16 V	8 ... 26 V	8 ... 16 V	8 ... 26 V
No-load speed	4800 rpm	4700 rpm	4800 rpm	4700 rpm
PWM-input (digital signal)	--		0 / 5 V, inverse PWM	
FG-output (Speed output)	--		6 pulses / revolution	
Inrush current (depending on series resistance and supply voltage)	34 A < 20 μ s @ 24 V, 0.7 Ω			
Input capacitor	14 A < 20 μ s @ 12 V, 0.7 Ω			
Thermal insulation class	10 μ F			
Protection class	A			
Protection class	IP30			
Electrical connection				
+Supply voltage	+VS	AWG 28	red	red
- Supply ground / GND	- VS / GND	AWG 28	black	black
Speed control input for inverse PWM control	PWM-input	AWG 28	--	blue
Speed output	FG-output	AWG 28	--	green
!	PWM control is not allowed on the +VS and -VS/GND terminals!			
	Reverse voltage is not allowed on the +VS and -VS/GND terminals!			
	Supply voltage is not allowed on PWM-input and FG-output!			

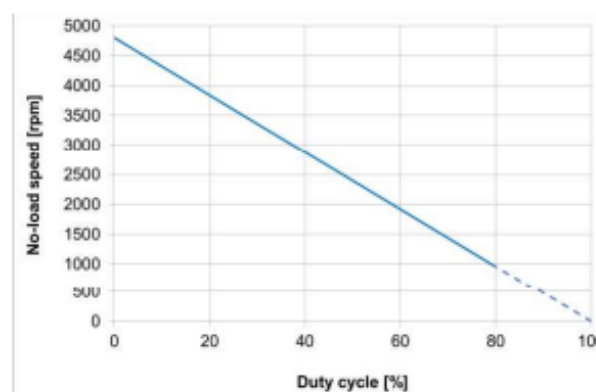
Auxiliary functions	
Motor current limitation	max. 0.75 A
Over-temperature protection electronics	
Blocking protection, no autonomous restart	
Under voltage shut down	≤ 8 V
Clear errors	Power cycle or PWM input signal 100 %, > 1 s

4-wire 12 V motor speed control with inverse PWM signal on PWM-input

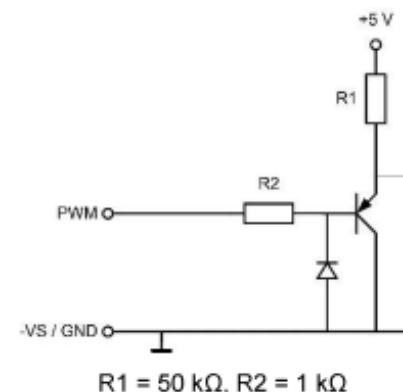


PWM control voltage level 'low'	$\leq 0,5 \text{ V}$	
PWM control voltage level 'high'	$2.7 \dots 5.5 \text{ V}$	
PWM frequency range	$15 \dots 25 \text{ kHz}$	
Duty factor for motor OFF	100 %	0 rpm
Duty factor for full speed	0 %	4800 rpm

Control voltage characteristics¹



PWM-input circuit



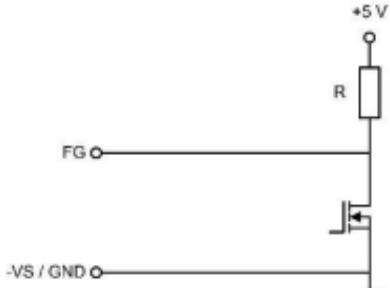
$R1 = 50 \text{ k}\Omega$, $R2 = 1 \text{ k}\Omega$

Connect PWM-input to -VS/GND

2 wire operating mode



Supply voltage is not allowed on PWM-input!

4-wire motor speed output on FG-output	
FG-output pulses	6 pulses / revolution
FG-output pulse time 'low'	50 %
FG-output level 'high' (no load)	5 ... 6 V
FG-output level 'low' (no load)	< 0.5 V
FG-output circuit	 $R = 10\text{ k}\Omega$
!	Supply voltage is not allowed on FG-output!